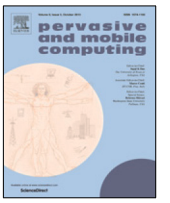
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Chirp-Loc: Multi-factor authentication via acoustically-generated location signatures



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ABSTRACT

Context-based authentication has been proposed as a way to enable secure authentication with minimal or even no user interaction requirements by using sensor data to ensure the device being authenticated is in possession of the person initiating the authentication request. A key limitation of practically all context-based authentication systems is that they are vulnerable to context manipulation attacks where an attacker manipulates the environment to create a desired response in the sensor data. We contribute Chirp-Loc as a system that has been designed to improve the robustness of context-based authentication solutions against context-manipulation attacks. Chirp-Loc integrates an innovative approach based on room impulse response (RIR) to establish a location fingerprint that characterizes the *physical environment* instead of the ambient environment. We describe the design and development of an Android prototype of Chirp-Loc. We also conduct extensive accuracy and security analysis of Chirp-Loc by considering a multi-factor authentication solution that uses Chirp-Loc to verify the proximity of an authentication token, such as a smartphone. Through extensive experiments, we demonstrate that Chirp-Loc offers high degree of security and usability. Our work paves the way for improving the resilience of context-based authentication against attackers that manipulate the context information and offers a way to implement authentication systems that minimize user interaction demands while offering a high degree of security.

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1. Introduction

Context-based authentication has been proposed as a mechanism for enabling secure authentication with minimal or even no user interaction requirements. The basic idea in context-based authentication is to use sensor data, such as audio, information about the wireless environments, or combinations of different sensors to verify that the authentication request comes from a person that is in proximity of the device being authenticated. Context-based authentication solutions are particularly well-suited for input constrained devices, such as XR devices [1] or emerging IoT devices that lack traditional means of input [2], or situations where the usability of the system suffers from a need of interaction.

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A key limitation of practically all context-based authentication schemes is their vulnerability to context manipulation attacks where an attacker manipulates the sensor data. For example, audio-based solutions can be manipulated by calling the user's phone and playing a specific high volume audio on the background [3] and wireless environment can similarly be manipulated by adding transmitting devices (e.g., phones in hotspot mode). The reason for this vulnerability comes from the fact that most solutions rely on sensor data that characterizes the user's *ambient environment* instead of the user's *physical environment*. As a result, and as will also be demonstrated in this paper for the case of audio, attackers can manipulate the characteristics of the ambient environment and break the authentication. To overcome this limitation, there is a need for a solution that can improve the resilience of context-based authentication systems.

We contribute *Chirp-Loc* as a novel system for improving the resilience of context-based authentication solutions against context-manipulation attacks. Chirp-Loc establishes an *unique signature of the physical environment* by capturing the room impulse response (RIR), the unique signature of sound reflections in the environment as a response to an audio stimulus (chirp). As this signature is a characteristic of the physical environment, Chirp-Loc overcomes context manipulation attacks unlike existing solutions that are based on characteristics of the ambient environment [4–7]. Chirp-Loc can also reliably defeat a proximity attacker who remains nearby the victim (e.g., several feet away) as location signatures differ based on the presence or absence of physical obstacles in between the source of the audio stimulus and the audio receiver. We describe the design and implementation of an Android prototype of Chirp-Loc. We also introduce a multi-factor authentication system that relies on Chirp-Loc and conduct extensive evaluations to assess the accuracy and resilience of Chirp-Loc against different attackers. The principle of using room RIR for co-presence detection has been presented in our previous work [8]. Chirp-Loc extends on our previous work by addressing challenges in integrating room impulse response into a full-fledged authentication system that generates, detects and analyzes RIR signatures. We also use our system to implement a working prototype of a multi-factor authentication solution for Android devices, and we conduct rigorous analysis of the accuracy and the security of the solution, extending on our previous work which only considered the co-presence detection accuracy of room impulse responses.

We evaluate Chirp-Loc through extensive experiments on data collected from five locations and three speaker location settings. The results show the signatures generated by Chirp-Loc to be highly localized and resilient against adversaries. Specifically, under a similar adversary setting, we show that prior audio-based context authentication schemes are entirely vulnerable (i.e., 100% success rate for adversaries) to proximity and remote attacks whereas Chirp-Loc can overcome these attacks in over 90% of the cases. While Chirp-Loc significantly improves the resilience of context-based authentication systems, the results also show that Chirp-Loc alone is not sufficient for authentication. The false acceptance rate (FAR) is comparable to other context-based authentication solutions (ranging from 4.1% to 9.7%), which is insufficient to be used as the sole authentication factor but highly effective when used in a multi-factor authentication context. The lowest FAR occurs in contexts with highest audio volume, making Chirp-Loc particularly effective at overcoming attacks that manipulate the ambient environment. We also briefly discuss how Chirp-Loc can be further improved by combining with sensors that characterize the ambient environment. Our work paves the way for improving the resilience of context-based authentication against attackers that manipulate the context information and offers a way to implement authentication systems that minimize user interaction demands while offering a high degree of security.

Summary of Contributions

- **Novel system**: Chirp-Loc is a novel system for characterizing the physical environment by establishing an unique signature based on room impulse responses. A prototype of Chirp-Loc has been implemented for Android smartphones.
- **Novel Multi-Factor Authentication Solution**: We present a multi-factor authentication solution that uses Chirp-Loc as one of the authentication factors.
- **Improved security**: Extensive experiments demonstrate Chirp-Loc to significantly improve the resilience of context-based authentication schemes against attackers, while providing reliable and accurate authentication.

2. Chirp-Loc Prototype System and Threat Model

Chirp-Loc is designed as a system that characterizes the user's physical environment and that supports context-based authentication. For simplicity, we focus on an authentication scheme where the sensor data is collected only during authentication request, but Chirp-Loc can also support continuous authentication by periodically triggering an authentication request and collecting sensor data to support it. Chirp-Loc can operate on any device that has a microphone and a speaker. Chirp-Loc leverages the physical characteristics of the location for authentication where the authentication protocol is executed to mitigate various types of attacks (presented in the next subsection). Replicating the location signature is highly difficult for an adversary as it depends on the physical characteristics of the environment where it is captured, making Chirp-Loc more resilient against attacks than current co-presence detection approaches. We next present the design and implementation of a multi-factor authentication scheme that relies on Chirp-Loc. The internal functions of Chirp-Loc are then discussed in the subsequent section.

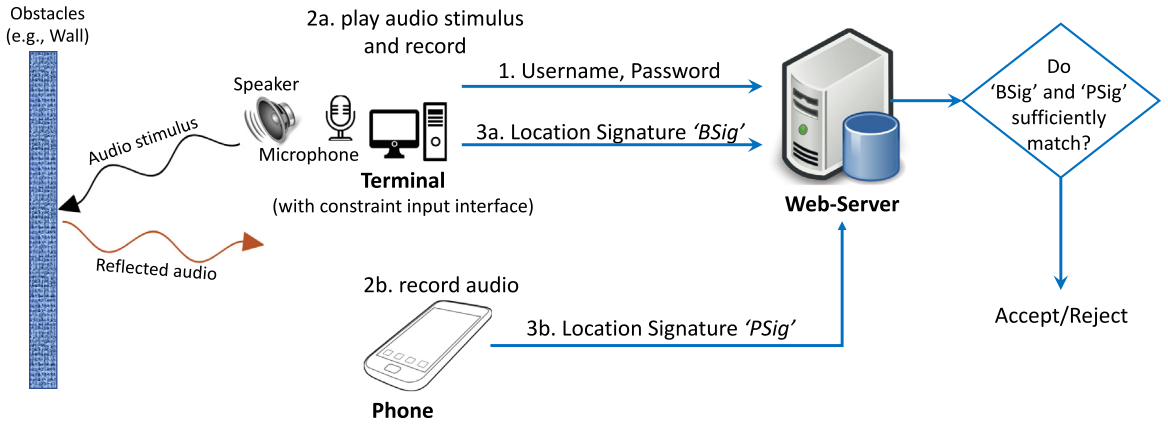


Fig. 1. Multi-factor authentication using Chirp-Loc. The user inputs the username and pass code to initiate authentication. The authenticating device then plays an active audio stimulus, e.g., a sine sweep sound, through the built-in or the connected speaker (e.g., wireless speaker, voice assistant device). The web server makes decision by correlating the location signatures captured by the device and the phone. The location signatures are derived from the room's impulse responses (RIRs) of the locations to the audio stimulus. The web server rejects the login if location signatures do not match or RIRs are not found.

2.1. Chirp-Loc Multi-Factor Authentication

We consider a system that requires the user to bring an authentication token (i.e., auxiliary device such as a smartphone used to support the authentication process) close to the authenticating terminal. Authentication is initiated by providing a username and a pass code. Next the server sends a short audio stimulus (a chirp in our current design) to the terminal. The terminal plays back the received chirp through the connected speaker. Both the terminal and the phone record the chirp and deconstruct the original chirp from the recording to extract the impulse response of the location where the chirp sound is recorded. The impulse response effectively captures the way the room modifies an audio stimulus and is solely dependent on the physical environment. As the impulse response is a function of the physical environment, it embodies a unique *location signature*. This contrasts with current co-presence solutions which instead focus on characterizing the ambient environment. The server correlates the pair of location signatures from the authenticating device and the phone to establish a measure of proximity between these two devices. Based on the similarity between the location signature pair, the server decides whether to accept or deny the login attempt. Fig. 1 shows a high-level overview of Chirp-Loc.

2.2. Threat model

We assume that an adversary can neither gain the physical access of nor compromise the authentication token, i.e., the phone. We also assume that the adversary cannot compromise the victim's authenticating terminal. If the adversary compromises the authenticating device, then he can easily bypass the security of Chirp-Loc, for example, by launching a man-in-the-middle attack and hijacking the victim's session with the remote server, thereby defeating any authentication mechanisms. Practically all schemes are vulnerable to such an attack, and hence, the attack is not within the scope of our threat model. However, our threat model still considers a strong adversary who has obtained the victim user's entry-point login credentials, i.e., his username and password, and attempts to access the victim's account on behalf of the user. The adversary can obtain the victim's credentials through phishing attacks, password database leakage, or other mechanisms [9–12].

We explore three types of adversaries – *Proximity Attacker*, *Remote Passive Attacker*, and *Remote Active Attacker*, as described below.

Proximity Attacker is a targeted and co-located attacker who attempts to log in to the victim's account while remaining close to the victim (or their phone). These are typically acquaintances, friends and co-workers, who turned into adversaries.

Remote Passive Attacker attempts to log in on behalf of the victim user when the victim is far away from the terminal without making any changes to the audio environment around the victim.

Remote Active Attacker tries to login on behalf of the victim user and deliberately attempts to make the environment around the victim (or his phone) and the attacker (or victim's terminal) similar. This type of attack has been explored as the *context-manipulation attack* in previous works [8,13].

When launching the attack against Chirp-Loc, a remote active attacker tries to make its environment similar to that of the victim's environment by ringing the victim's phone (through calls or messages) and generating a similar ringtone at his end (i.e., near the authenticating terminal). Since our threat model assumes that the adversary already knows the victim

user's phone number and his phone's ringtone, he can launch the attack against Chirp-Loc following the steps similar to that of the Sound-Danger attack system [14]. The adversary also can mute the stimuli played through the speaker to maximize its chances to succeed. The remote active attacker could also launch the *same-media attack* as suggested in Sound-Proof [4]. Relay attack is considered as one type of remote active adversary.

3. Processing pipeline and prototype implementation

Our prototype implementation of Chirp-Loc has four core components – (1) Browser and Web-Server, (2) Phone Application, (3) Chirp Detector, and (4) Location Signature Extractor – as described below.

3.1. Chirp detection

Chirp-Loc uses a chirp sound as the audio stimulus for generating a signature of a physical location. The first step is to detect the presence of the chirp. To accommodate device heterogeneity and its sampling rate (f_s), we resample the recorded audio to match the sampling rate used for the excitation signal. In case the recording has more than one audio channel, we use the first channel and drop the other channels. In our experiments, we assume the sampling rate to be 44100 Hz, which is the most common frequency for audio.

After the sample rates have been equalized, we reduce background noise. We use 800 Hz as cut-off frequency as this covers both common low frequency noise as well as the lower end of the mid-range noise spectrum. This can be accomplished using a high-pass filter on the recording. We apply the filter on a separate copy of the signal and use it solely for identifying the chirp, whereas location signatures are computed from the original signal.

After the filtering, we center and scale both the recorded and the excitation signal using z-score normalization, and compared the two samples using cross-correlation. If the maximum correlation score is above a threshold (empirically chosen), we declare that the chirp is present. In the experiments, we use an empirical value corresponding to 10% of the perfect alignment (i.e., $f_s \cdot 0.1$). All recordings where no chirp is detected are omitted to save resources and to reduce the range of potential false positives. Listing 1 presents the pseudocode of our implementation of the chirp detection module.

```
1   if not fs(signal) == fs(recording)
2     recording = resample(recording, fs(signal))
3
4   recording = highpass(recording, 800)
5   recording = zscore(recording)
6   signal    = zscore(signal)
7
8   if max(abs(xcorr(signal, recording))) < 4000
9     return false /* chirp is not detected */
10  else
11    return true /* chirp is detected */
12  end if
13 end function
```

Listing 1: Chirp Detection Algorithm

3.2. Location Signature Extractor

Location Signature Extractor is responsible for generating physical features of the space where the user is located. Chirp-Loc measures the features via the reflection of the space to an audio stimulus (a chirp). Location Signature Extractor creates the location signature of the supplied audio sample. Location signature is a set of RIR features, reverberation time (RT60), early decay time (EDT), early-to-late ratio, and direct-to-reverberant ratio, extracted by first computing RIR from recorded audio signals and then selecting its features (using a machine learning technique). We adopt a similar technique presented in [8] in our implementation to compute RIR. The web server compares the similarity of location signature pair derived from the two devices under considerations (by using a pre-trained machine learning model detailed in Section 5.2.1) to make accept or deny access decisions.

3.3. Browser and Web-Server

The browser application (implemented using HTML, JavaScript, and CSS) has a simple button to control the audio recordings. Chirp-Loc uses GCM (Google Cloud Messaging) to communicate with the designated Android phone. When the button is pressed, the browser application triggers the audio recordings on the browser and the phone. It also plays a sine sweep audio clip as an audio stimulus simultaneously while it is recording audio. The purpose of recording at the browser side is to simulate a scenario where authenticating device plays the audio stimulus via a connected speaker and records it, i.e., the second factor in a multi-factor authentication scheme. Specifically, the main purpose of browser recording is to bind the distance between the phone and the browser during a login attempt through RIR similarity. To record audio through the browser, Chirp-Loc uses HTML5 WebRTC API [15]. The recordings from the browser are uploaded

to the web server for the purpose of offline analysis in our current implementation. In a real-world implementation, the authenticating device (or an application of Chirp-Loc) would extract the RIR features from the recorded audio and transmit it to the web server. The web server then processes the RIR features, applies a pre-trained classification model to decide the proximity of devices, and make an authentication verdict.

3.4. Phone application

We have implemented Chirp-Loc multi-factor authentication solution as an Android app for the phone that stays idle in the background. The phone app is automatically activated when a GCM signal arrives from the web server, and it starts recording. As soon as the recording is completed, the audio recording is stored locally on the phone for the purpose of offline analysis. In the real-world implementation, Chirp-Loc transmits RIR features extracted from the recorded audio on the phone to the web server for further analysis.

4. Experiment setup

We evaluate Chirp-Loc and the multi-factor authentication prototype in extensive experiments. We use a Thinkpad (Intel Core i7 2.6 GHz) as a login terminal (emulating an input constrained terminal device), Sony SRX XB2 [16] as the external speaker, and Samsung Galaxy S6 smartphone (running Android 6.0.1) as the authentication token. Different from the work in [8], we assume our authentication system is designed and trained for a specific pair of token-terminal. The concern of whether the location signatures are sufficient with various device models and in many different locations have been addressed and proved in [8]. Our experiment focuses on one pair of terminal and token with various settings of positions and sound volumes. We also consider an extensive security analysis against different types of attackers. We investigate three important factors that have a significant effect on the authentication decision of the Chirp-Loc system: the *distance* of the phone from the terminal (the terminal device in a specific setting), the *volume level* at which the terminal plays back the audio stimulus via the speaker and the *location* of the speaker.

Distance: We collected audio samples by positioning the phone at six different distances from the terminal. Specifically, we positioned the phone at a distance of 0 ft (i.e., close to the terminal) to 6 ft from the terminal with an increment of 1 foot. We categorized these six distances into the following two groups.

- *Benign Distance:* As noted earlier, Chirp-Loc limits the access to authenticating devices to 3 ft (i.e., personal space), and therefore requires the user to bring his phone nearby the authenticating the device during the authentication process. Chirp-Loc assumes that the user keeps his phone close by at a distance less than or equal to 3 ft. Such distance settings are considered as the *benign* distance in our study.
- *Attack Distance:* People known to the user, such as neighbors, friends, and co-workers, may typically remain at a distance ranging from 4–5 ft (i.e., social distance). This represents an easy and relaxed space for talking, shaking hands, and gesturing. Such known persons may turn into adversaries and may attempt to login on behalf of the user. We consider more than 3 ft as the distance between the terminal device and the user's phone, termed it as the *attack* distance.

Volume: Since each user may have their personal preference towards the volume level of the terminal and that of the speaker, we consider three different volume levels in our evaluation: (a) *Full Volume*, (b) *Medium Volume*, and (c) *Low Volume*. In the full volume setting, the maximum loudness of the sound perceived at the terminal when the chirp sound is played through the speaker is 80 dBA while it is 75 dBA and 70 dBA for medium and low volume settings, respectively. We used Digital Sound Level Meter to measure the loudness of the terminal at each of these volume settings. We note that most users typically set the audio volume between (75–105) dBA [17].

Speaker Location: As each user may have their personal preference towards positioning the speaker in a room, we consider three different location settings. We note that the distance of the speaker from the terminal does not have any effect on the performance of Chirp-Loc, but the loudness of the chirp sound perceived at the terminal may affect its performance. The three settings are:

- *Below-Desk:* The user may place the speaker below the desk where he generally uses his personal computing device (e.g., desktop PC, laptop). We put the speaker underneath the desk on the floor.
- *Room-Corner:* Some users may also prefer putting the speaker at one of the corners of the room for better sound effects. We place the speaker at one of the corners of the experimental room. In this setting, the distance between the terminal and the speaker was approximately 6.6 ft ($\simeq$2 m) on average.
- *On-Desk*: The user may also keep the speaker on the desk where he uses his terminal. We keep the speaker on the desk at approximately 2 ft behind the terminal.

Naturally considering all possible types of rooms (i.e., different shapes, sizes, and artifacts) is not feasible and instead we design our evaluation to capture six different locations that are representative of common everyday locations — *office* setting where people were working in their terminals, *conference room* where people have their meeting, *classroom* setting where the students at the university take classes, *living room*, and *bedroom* for the home (private) environment. We note

Table 1
Data samples for various settings.

Experiments	No. benign pairs	No. attack pairs
All volume levels		
Below-Desk	268	241651
Room-Corner	254	216425
On-Desk	223	160202
Low volume setting		
Below-Desk	88	26700
Room-Corner	90	27300
On-Desk	75	17880
Medium volume setting		
Below-Desk	90	26850
Room-Corner	89	26551
On-Desk	74	17641
High volume setting		
Below-Desk	90	27000
Room-Corner	75	18250
On-Desk	74	17880

that many of these locations (e.g., office room, conference room, classroom) are chosen from the same building and hence show certain homogeneity (as expected from different rooms of the same building). At each of these locations, we collected audio samples considering the different combination of distance, speaker location, and its volume level settings.

Datasets. We first collected an attack dataset for the passive attack scenario. In an office room, we collected 300 attack samples representing various attacks mentioned in our threat model (as described in Section 2.2). These attack samples do not capture RIR, yet our Chirp-Loc system, specifically Chirp Detector, should be able to detect these adversarial samples based on the lack of chirp.

Our data collection results in three datasets *Below-Desk*, *Room-Corner*, and *On-Desk*. For each dataset, samples were collected with various volume and distance settings at six different locations (as described earlier). In sum, we obtained 1077 samples for Below-Desk, 1019 samples for Room-Corner, and 895 samples for On-Desk. From these samples, we form copresence and non-copresence paired samples. Similar to previous work [18,19], we use the collected audio samples to emulate benign as well as adversarial executions of the protocol. A benign execution of the protocol is one where the two devices are truly copresent. An adversarial execution happens when the two devices are not copresent. Any pair of samples recorded by two distinct devices (where one plays an audio clip) at the same time, in the same room and withing benign distance is considered a benign pair. Any pair of samples recorded by two distinct devices (where one also plays an audio clip) in otherwise settings (different rooms, different times, in attack distance) is considered an attack pair. Details of datasets are shown in Table 1. The numbers of benign pairs are small as it requires an exact match of two devices' settings; while the numbers of attack pairs are large due to the combinations of many different devices' settings.

5. Security analysis

We next conduct security and usability analysis of Chirp-Loc, focusing on the three different attacks as described in our threat model. The co-presence detection performance is evaluated in our previous work [8] and is approximately 90% across different environments. We also detail the machine learning methods that are used and demonstrate that the performance of Chirp-Loc is robust to different device positions, volumes, and chirp signal lengths. We end the section with a brief security comparison of Chirp-Loc and current state-of-the-art solutions.

5.1. Defense against remote passive attack

A remote passive attacker does not manipulate acoustic environments at the targeted victim's ends. It simply allows the authenticating terminal to play a chirp audio clip while the phone is in a remote location. Since the phone is away, it cannot capture the chirp played at the terminal side, thus its recording does not contain the expected chirp emitted by the terminal. By checking whether the chirp was recorded by both the phone and the terminal device, the Chirp-Loc system can detect the attack.

We tested our Chirp-Loc system, specifically the Chirp Detector, against 300 passive attack samples. Chirp-Loc was able to correctly identify 299 samples that do not contain the expected chirp signal. It failed to identify only one such sample because it contains a random background noise that was similar to the exciting chirp signal.

Table 2
Error rates for various settings (Distances, Volumes and speaker Locations).

		Below-Desk		Room-Corner		On-Desk	
		Prediction					
		1	**0**	**1**	**0**	**1**	**0**
All volume levels							
GT	**1**	242	26	242	12	211	12
	0	14447	227204	15336	201089	11612	148590
FRR		0.097		0.047		0.054	
FAR		0.06		0.071		0.072	
Low volume setting							
GT	**1**	81	7	83	7	68	7
	0	2584	24116	2591	24709	1448	16432
FRR		0.08		0.078		0.093	
FAR		0.097		0.095		0.081	
Medium volume setting							
GT	**1**	83	7	86	3	70	4
	0	1972	24878	2456	24095	1374	16267
FRR		0.078		0.034		0.054	
FAR		0.073		0.093		0.078	
High volume setting							
GT	**1**	86	4	71	4	69	5
	0	1597	25403	1031	17219	736	17144
FRR		0.044		0.053		0.068	
FAR		0.059		0.056		0.041	

5.2. Defense against remote active attack

Remote active attack is the strongest attack in our threat model. We next demonstrate how Chirp-Loc is robust against this attack.

5.2.1. Audio manipulation attack

Remote active attackers in our threat model have the capability of manipulating the audio environment by injecting the same audio at both victim sides (e.g. Sound-Danger attack [14] and same-media-attack in Sound-Proof [4]). If attackers play arbitrary audio rather than the required chirp audio, the Chirp Detector can detect the absence of the chirp and the Location Signature Extractor is not able to generate location signature and thus the web server refuses access as the authenticating factor (i.e., presence of chirp audio) requirement is not fulfilled.

In case there is a presence of strong active attackers who can exactly inject the same chirp audio at both victim sides, we conduct experiments to simulate these strong remote active attackers. In our experiment, the same chirp signal is played at different locations. Chirp-Loc uses a machine learning technique to provide a verdict on the copresence of the terminal device and the phone. We consider copresence verification as a classification task for labeling two classes: *copresent* (positive class — 1) and *non-copresent* (negative class — 0). Ground truth (GT) is labeled based on room identifiers that we noted during data collection. We selected the RandomForest classifier which performs best according to [8].

The way we generated benign and attack pairs makes our dataset heavily imbalanced towards attack pairs. To overcome the imbalance issue for the classifier, we employ undersampling, and specifically the RandomUnderSample of Scikit-learn library, on the set of attack pairs during the training phase. We do not alter the imbalance of the attack pairs during the testing phase. Undersampling ensures the distribution of the benign pairs remains intact. Indeed, the high number of combinations between attack pairs means the majority class contains higher variation which makes it safer to reduce samples from it rather than duplicate those in the minority class (i.e., oversampling). Undersampling has also been shown to be better for the classifiers we use in our experiments [20].

Similar to [8], for each sample pair (of both benign and attack pairs), using their RIR acoustical feature vectors, we compute (component-wise) the vector of squared differences, yielding a 244-elements feature vector. We then applied the RandomForestRegressor algorithm implemented in Scikit-learn to select the 50 most relevant features.

To evaluate our classification model, we used a 10-fold cross-validation approach: for each dataset, we randomly divided the data into 10 subsets (folds); 9 subsets were used for training the model and the remaining subset for testing. We use 10-fold cross-validation as it offers a robust estimator of the performance metrics, while aligning with the process that was used in our earlier work [8], thus making comparison to our prior work easier. We present the performance of Chirp-Loc with different volume settings. The performance of Chirp-Loc is measured by FRR and FAR (see Section 2.1). Table 2 reports the confusion matrix, FRRs, and FARs. A detailed explanation of the results comes below.

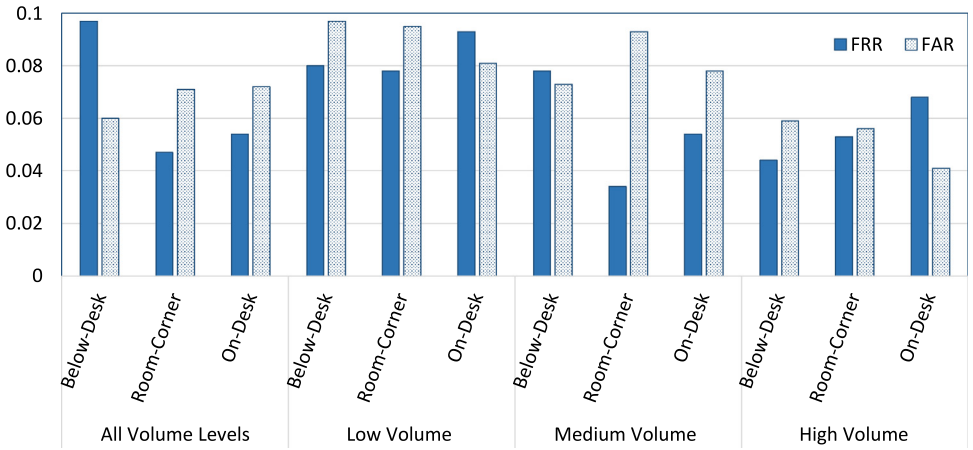


Fig. 2. Error Rates (i.e., FRR and FAR) with various volume settings.

All volume levels. In this setting, audio samples collected with any level of volume that the browser perceives are taken into account. That is, samples are paired based on the room and the distance between the browser and the phone, regardless of the volume level of the browser. Copresent pairs represent the benign case and they have similar volume levels as they were recorded in the same session. Non-copresent pairs (attack cases) combine samples with the same or different volume levels. Notably, FRR can be as low as 0.047. That is, when the two devices are copresent, Chirp-Loc fails to detect copresence roughly 5% of the time. Also, FAR can be as low as 0.060. That is, in the presence of an adversary, Chirp-Loc fails to detect the attack roughly 6% of the time.

Low volume setting. In this setting, we consider only samples recorded with the low volume setting (70 dBA). The FRR can be as low as 0.078, and FAR can be as low as 0.081. When comparing this result with the all volume levels setting, we can see that the performance is slightly reduced (higher error rate). This can be explained by the loss of audio energy recorded at a low volume level. In such situations, the performance can be enhanced by requiring a minimum threshold on the audio energy and to instruct the user to bring the device closer to the terminal before the authentication can be accepted.

Medium volume setting. Here, we consider only samples that were recorded with the medium volume setting (75 dBA). Compared to the low volume setting, we can observe the increment of the performance (lower error rate). The FRR can be as low as 0.034 and the FAR can be as low as 0.073.

High volume setting. With the high volume setting, Chirp-Loc obtains a low error rate. FRR can be as low as 0.044 and FAR can be as low as 0.041. This setting obtains the best results among the others. Clearly, the higher the volume of the sound, the better the perceived RIR signals, thus better RIR features, which are useful for the classifier.

Fig. 2 depicts the performance of three datasets with varying volume levels. In all datasets, FARs are lower than 0.1. This result shows the security offered by the Chirp-Loc system. Given the presence of remote active attackers, their chance of success is lower than 10%. While security is a critical feature of Chirp-Loc, the functionality of the system is not downgraded and it can still be as low as of 0.034. This RIR features based model defenses the system against remote active attackers including Sound-Danger attacks and Same-Media attacks. Note that other co-presence based authentication schemes, such as Sound-Proof [4], a popular and closely related approach, are vulnerable to this remote active attacker.

5.2.2. Location signature manipulation attack

In the previous section, we showed that Chirp-Loc, which uses acoustically-generated location signatures, can defend the authentication against the audio manipulation attack. In that setting, we assume that location signature is unique to a specific location and it is not easy to manipulate as how attackers can do to alter the audio environment. A possible concern to the security of Chirp-Loc is about the hardness of reproducing location signature. In our work, reproducing the location signature means to create a space that generates a similar RIR to an existing RIR. Up to the time of this writing, we are not aware of any available method to construct a space given a specific RIR.

5.3. Defense against proximity attack

Proximity attack refers to the setting where the attacker is in close proximity to devices (i.e., the terminal and the phone). We assume that Chirp-Loc is able to detect such attacks given a predefined distance for copresence. To do so, we

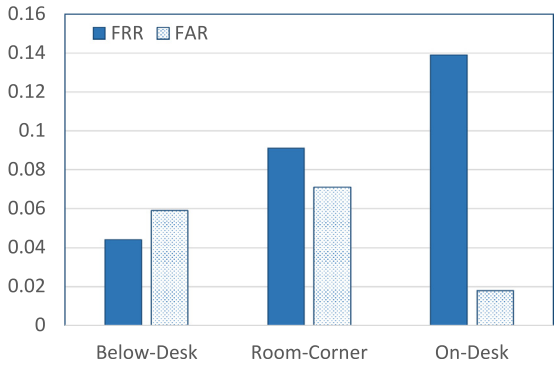


Fig. 3. Error rates for proximity attack.

Table 3
Error rates for proximity attack.

		Below-Desk		Room-Corner		On-Desk	
		Prediction					
		1	**0**	**1**	**0**	**1**	**0**
GT	**1**	86	4	231	23	192	31
	0	1597	25403	18	237	4	219
FRR		0.044		0.091		0.139	
FAR		0.059		0.071		0.018	

Table 4
Data samples for various Chirp Lengths.

Experiments	No. benign pairs	No. attack pairs
500 ms	49	1360
1 s	50	1400
2 s	50	1400

train a classification model such that it can distinguish pairs of samples recorded within a given distance of devices in the same room with other pairs of samples in the same room but recorded at a larger distance. We filter our data samples to obtain only pairs of samples in the same room. We choose the average distance of 3 ft as a cut-off point. That is, any pair with a distance less than 3 ft is considered a benign sample and the rest of samples with larger distance ($>$ 3 ft) is labeled as attack sample. We show the classification result over three datasets in Table 3 and Fig. 3. Given the result, we can see that attackers only succeed with lower than 10% in two datasets, and 14% with one dataset. Other proximity-based authentication schemes such as [4,5] are either entirely vulnerable to proximity attack or can hardly prevent it.

5.4. Sufficiency of chirp length

In our experiment, Chirp-Loc uses a short chirp of 2 s length to excite the acoustical spaces. The reason behind this chosen length is to cover a large range of spaces, from small rooms such as offices, homes, and classrooms to large public spaces like church or auditorium. Based on the application context, chirp length can be shortened to 1 s or 500 ms. Our data collected in small spaces results in an RIR signal with a length smaller than 1 s. For such spaces, very short chirp duration is sufficient and it results in faster recordings for RIR. To evaluate this claim, we conducted a small scale experiment on RIR samples collected and extracted in 5 small-size rooms, with distances between devices 1 ft and 2 ft, volume 100% (see data details in Table 4). Results as shown in Table 5 and Fig. 4 confirm our claim that very short chirp length 500 ms or 1s is efficient for small-size spaces.

5.5. Comparison to audio proximity based approaches

Table 6 compares Chirp-Loc to existing approaches highlighting how it is only solution that is resilient to multiple different attacks. Chirp-Loc is independent of the ambient environment and significantly more robust against three different attacks. For details of the different solutions, please see Section 7.

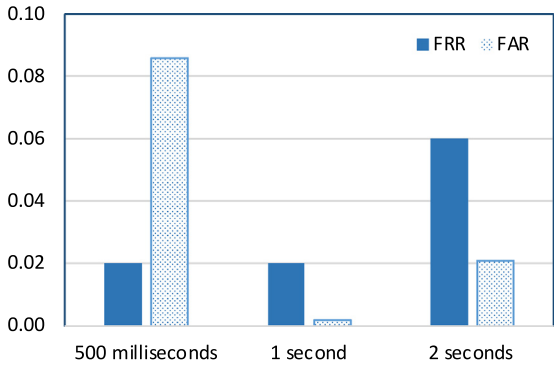


Fig. 4. Error rates for various chirp lengths.

Table 5
Error rates for various Chirp Lengths.

		500 milliseconds		1 second		2 seconds	
		Prediction					
		1	**0**	**1**	**0**	**1**	**0**
GT	**1**	48	1	49	1	47	3
	0	117	1243	0	1400	30	1370
FRR		0.02		0.02		0.06	
FAR		0.086		0.0		0.021	

Table 6
Comparison of Chirp-Loc to existing audio proximity based authentication approaches. Key criteria are audio ambience independence, resilience to proxity (Attack I), remote passive (Attack II) and remote active (Attack III) attacks. Notion: ✓yes, ✗no, ? no information.

Approaches	Audio independence	Attack I	Attack II	Attack III
Sound-Proof [4]	✗	✗	?	✗
Listening-Watch [5]	✓	?	?	✗
Contextual ZIA [18]	✗	✗	✗	✗
DoubleEcho [8]	✓	?	✓	✓
Chirp-Loc	✓	✓	✓	✓

6. Discussion

Naturally there is room for further work and improvements. We discuss a few points here.

Outdoor environments. Whether RIR is sufficient to capture location signature in outdoor environments is still an open question. Although RIR is captured only in a closure space where a sound is reflected and obstacles make it travel in multiple paths creating a unique location signature, we still foresee the ability to capture RIR in an outdoor environment. However, it might require a careful selection of parameters for the sound source and the location of the devices involved. A rigorous future study is needed to study the feasibility of using RIR in outdoor environments for authentication.

Potential Improvements. The main limitation of Chirp-Loc currently is the FAR which, while reasonable, is unfortunately too high to be used as the sole authentication factor. The main reason for this is that some background audio signs can generate audio energy on multiple frequencies, making the signals resemble the chirp. This could be potentially mitigated using the lower frequencies (i.e., those below 800 Hz) for establishing an ambient fingerprinting and combining this with the physical signature offered by Chirp-Loc. Another option would be to dynamically adjust the frequency range where the chirp is played based on a short recording of the ambient noise. Our results showed the potential of using RIR as continuous signature of the physical environment and these extensions can be used for further improving the performance of Chirp-Loc.

Usability. The advantage of using chirp sound in Chirp-Loc is that it alerts the users about the ongoing authentication process, thanks to the generated sound by the terminal. Unfortunately, the wide-band chirp might be uncomfortable for users. Several approaches can be employed to improve the user experience. The chirp length can be reduced to as short

Table 7
Comparison of different approaches.

Approach	Brief description	Vulnerability
Proximity-based	Detect proximity between user and device being authenticated using secure token.	Vulnerable to man-in-the-middle and relay attacks, does not require intent to authenticate
Audio-Based Proximity Detection	Use ambient audio to verify proximity	Susceptible to context attacks where the audio environment is manipulated.
Active audio	Play an active audio signal that needs to be captured on secure token.	Vulnerable to relay and replay attacks.
Context-based Proximity Detection	Use context information from different modalities to verify proximity.	Vulnerable to attacks that manipulate the signal environment, need same sensors on all devices.
Physical and Behavioral Biometrics	Measure user characteristics or behavior and use that for authentication.	Require specialized hardware or activities from the user.
Our work	Active-audio based estimation of physical characteristics of the environment.	Vulnerable to relay attacks if physical environment can be controlled.

as possible, which can be 500 ms, as we argued above. An inaudible sound, instead of the audible chirp sound, can be used in Chirp-Loc, but for this, we have to wait until the recording capacity of commodity devices get advanced.

Privacy. Using audio has been a long debate when it comes to privacy concerns. Chirp-Loc uses chirp as a proxy to capture the location signature. The recorded audio is not used directly for colocation comparison and it is also not sent to any party even between the two entities — the authenticating device and the remote server. Only RIR features such as RT60, EDT, and other reverberation parameters, which are extracted by using the original chirp and the recorded audio, are used for authentication. It is not possible to revert RIR features to the original recorded audio. Therefore, ambient privacy is preserved.

7. Related work

Chirp-Loc is an example of a *contextual authentication solution* which uses context information to verify that the user is close to the device that should be unlocked. The earliest such works relied on simple proximity detection which can be broken by relaying user credentials or masquerading another device as the security token carried by the user. The main alternative is to integrate a context sensing layer to verify that the user is indeed in proximity of the device being authenticated to. Most of these solutions have relied on audio data, but also other sensor modalities, such as wireless signals, have been harnessed. The key problem with these approaches is that they break if an attacker can control the context information (audio or wireless). Chirp-Loc differs from these approaches by using physical characteristics of the environment to detect proximity. Below we discuss the main related works and Table 7 compares our approach to existing techniques.

Proximity-based approaches authenticate the user by detecting his/her proximity to a device. Examples of these systems include ZIA [21] and commercial solutions such as Apple's Auto Unlock [22] and Atama's Sesame [23]. These solutions are based on proximity detection of a token, such as an additional device, and only determine when the user is near to the device and ignore whether the user has an actual intent to use the device. Thus, these solutions can result in unintentional authentication. Another issue with these solutions is that if the attacker can masquerade as the device that is used as security token (e.g., by faking or relaying the network identifier) the system can be broken. Our work overcomes this limitation by ensuring the user and the device are in the same usage context.

Audio-based proximity detection is the most common approach for implementing two-factor authentication on input-constrained devices. The idea is to compare the audio environment of the security token with that of the device being authenticated and only approve the authentication whenever the audio environment is sufficiently similar. The best know example of these techniques is Sound-Proof [4]. The main limitation of these techniques is that they are susceptible to context attacks where the attacker takes control of the audio environment, e.g., by playing a specific sound [3,14]. Chirp-Loc overcomes this issue by linking the authentication to the physical characteristics of the environment, making the audio information difficult to control or to manipulate.

Active audio solutions use active audio to generate a signal that must be captured on the security token and verified to enable authentication. For example, SoundAuth [6] uses a randomly generated near-ultrasound signal (i.e., 18–20 kHz) to issue a challenge that needs to be accurately captured on the security token to allow authentication. Listening Watch [5] is a related approach that encodes a code word containing a random challenge in a browser generated human

speech. A wearable device acting as the security token needs to capture this audio and decode the code word to enable authentication. These approaches are vulnerable to a relay attack where one attacker triggers the authentication (by masquerading as the security token) and relays audio information to a second attacker who is located close to the security token and replays it.

DoubleEcho [8] is a generic solution that uses active audio to estimate physical characteristics of the environment and to establish copresence. This is accomplished by using an active signal to estimate the room impulse response which captures the way the room changes a given signal. As the room impulse response is a function of the physical environment, DoubleEcho is robust against attacks that manipulate the audio environment. Chirp-Loc relies on DoubleEcho as the proximity verification technique, but extends on it in several ways to enable robust authentication. Specifically, unlike DoubleEcho which assumes the presence of active audio for room impulse response estimation, Chirp-Loc separately analyses the audio environment to detect the presence of the active audio signal. This reduces the risk of false positives and decreases the risk of false authentication. Chirp-Loc also extends DoubleEcho to be resilient against proximity attacks.

Context-based proximity detection approaches rely on different context information to detect copresence. The audio solutions discussed above are examples of these, but also other modalities have been explored. For example, Next2You [24] uses Wi-Fi channel state information as context. Haus et al. [25] uses multimodal, Wi-Fi, Bluetooth, and GPS with various spatio-temporal granularity as context. Conti and Lal [26] presented a systematic survey on context-based copresence detection techniques. These approaches are all different from Chirp-Loc as it is not reliant on ambient environment but physical characteristic of space. The main limitation of all these approaches it that they are vulnerable to attacks that manipulate the context information.

Biometric schemes are another popular authentication solution. For instance, Fingerprint and FaceID have been widely deployed in mobile devices. Heart-related unique information, such as electrocardiogram (ECG) [27], photoplethysmogram (PPG) [28,29], and signal pulse response at the hand palm [30], has been suggested for user identification. Chirp-Loc relies on microphones and speakers which are available on most off-the-shelf devices, whereas these approaches require the necessary sensors to be integrated on the authentication devices.

Other Approaches: Pet2Auth [31] authenticates by having the user performs a physical operation (e.g., twisting, clicking, or swiping) on the device, and correlating events observed on the device and wrist-movements captured by a wrist-worn device. Chirp-Loc neither requires the user to wear a wrist-device nor perform physical operations, unlike Pet-2-Auth, Move2Auth [32] and SPIRE [33] authenticate the user by having the user moves his smartphone around the authenticating device and comparing the changes in Received Signal Strength (RSS) with the smartphone sensor. RSS is not reliable as it may be interfered with and affected by the presence of multiple devices. RSS is also susceptible to relay attacks [31]. TouchAuth [34] authenticates a user based on induced body electric potentials (iBEPs) generated when touching an analog-to-digital converter (ADC) pin on the authenticating device. It requires the user to wear a customized wrist-worn device during the authentication process. Further, it needs hardware changes on the device to expose an ADC pin. Unlike TouchAuth, our approach neither requires a customized wrist device nor hardware changes.

8. Conclusion

We presented Chirp-Loc as a novel system for improving the resilience of context-based authentication solutions. Chirp-Loc is based on the notion of acoustically-generated *location signatures* derived from the room's impulse responses. At its core, Chirp-Loc uses a machine learning technique to correlate location signatures pair captured by the phone and the authenticating device to determine their proximity. Unlike prior context-based schemes, Chirp-Loc is resistant to proximity and remote attacks since the system relies on the physical characteristics of the location which are hard to replicate. We presented the design and implementation of Chirp-Loc, including an Android prototype for multi-factor authentication. Evaluation results from experiments conducted under various parameter settings, showed that Chirp-Loc, using active chirp sound, enhances the security of authentication system (by offering a low false acceptance rate) without imposing much burden on the user by offering a low false rejection rate and fast sensing process.

Declaration of competing interest

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Data availability

Data will be made available on request.

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